INCH-POUND
MIL-DTL-52471/1D
30 September 2019
SUPERSEDING
MIL-DTL-52471/1C
4 December 2008

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLIES, RUBBER, HYDRAULIC PRESSURE-TYPE, TYPE 100R1, SINGLE- WIRE BRAIDED REINFORCEMENT HOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52471.

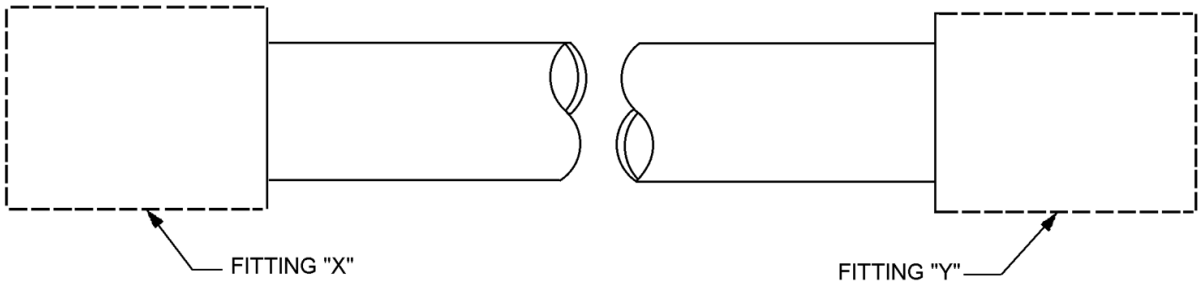


FIGURE 1. Hose assembly, fitting identification.

Operating pressure. Hose assembly shall operate at pressures shown in table I.

TABLE I. Hose assembly operating pressure. 1/ 2/

Dash number	Hose ID			Operating pressure max psi (MPa)
	Inch fraction	Inch decimal	mm	
-4	1/4	.250	6.35	2750 (19.0)
-6	3/8	.375	9.53	2250 (15.5)
-8	1/2	.500	12.70	2000 (13.8)
-12	3/4	.750	19.05	1250 (8.6)
-16	1	1.000	25.40	1000 (6.9)
-20	1-1/4	1.125	28.58	625 (4.3)

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

Hose assembly code number. The purpose of this slash sheet is to provide hose assembly code numbers for determining the hose diameter, fitting X, fitting Y, and bend angles (when applicable) as shown in tables II through VII. The bend angles are called “style codes” and consist of hose assembly styles as explained in MIL-DTL-52471.

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TABLE II. 1/4-inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN 1/	Fitting X (see figure 1)		Fitting Y (see figure 1)		Assembly code number 3/
		PIN	Bend angle 2/	PIN	Bend angle 2/	
1/4 .250 (6.35)	M52471/4-4	M52525/5-4-4	0°	M52525/5-4-4 M52525/5-4-5	0°	A001 A002
				M52525/6-4-4 M52525/6-4-5	45°	A004 A005
				M52525/7-4-4 M52525/7-4-5	90° 4/	A006 A007
		M52525/5-4-5	0°	M52525/8-4-4 M52525/8-4-5	90° 5/	A008 A009
				M52525/5-4-5	0°	A010
				M52525/6-4-4 M52525/6-4-5	45°	A012 A013
				M52525/7-4-4 M52525/7-4-5	90° 4/	A014 A015
				M52525/8-4-4 M52525/8-4-5	90° 5/	A016 A017
		M52525/6-4-4	45°	M52525/6-4-4 M52525/6-4-5	45°	A025 A026
				M52525/7-4-4 M52525/7-4-5	90° 4/	A027 A028
				M52525/8-4-4 M52525/8-4-5	90° 5/	A029 A030
		M52525/6-4-5	45°	M52525/6-4-5	45°	A031
				M52525/7-4-4 M52525/7-4-5	90° 4/	A032 A033
				M52525/8-4-4 M52525/8-4-5	90° 5/	A034 A035
		M52525/7-4-4	90° 4/	M52525/7-4-6 M52525/7-4-5	90° 4/	A036 A037
				M52525/8-4-4 M52525/8-4-5	90° 5/	A038 A039
		M52525/7-4-5	90° 4/	M52525/7-4-5	90° 4/	A040
				M52525/8-4-4 M52525/8-4-5	90° 5/	A041 A042
		M52525/8-4-4	90° 5/	M52525/8-4-4 M52525/8-4-5	90° 5/	A043 A044
		M52525/8-4-5	90° 5/	M52525/8-4-5	90° 5/	A045

1/ Part or Identifying Number (PIN).
2/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
3/ A suffix indicating plating finish shall be specified with the assembly code number, see table VIII.
4/ 90° short drop fitting.
5/ 90° long drop fitting.

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TABLE III. 3/8-inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN	Fitting X (see figure 1)		Fitting Y (see figure 1)		Assembly code number 2/
		PIN	Bend angle 1/	PIN	Bend angle 1/	
3/8 .375 (9.53)	M52471/4-6	M52525/5-6-6	0°	M52525/5-6-6 M52525/6-6-6 M52525/7-6-6 M52525/8-6-6 M52525/6-6-6	0° 45° 90° 3/ 90° 4/ 45°	A201 A203 A205 A207 A216
		M52525/6-6-6	45°	M52525/7-6-6 M52525/8-6-6	90° 3/ 90° 4/	A218 A220
		M52525/7-6-6	90° 3/	M52525/7-6-6 M52525/8-6-6	90° 3/ 90° 4/	A227 A229
		M52525/8-6-6	90° 4/	M52525/8-6-6	90° 4/	A234

1/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
2/ A suffix indicating plating finish shall be specified with the assembly code number, see table VIII.
3/ 90° short drop fitting.
4/ 90° long drop fitting.

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TABLE IV. 1/2-inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN	Fitting X (see figure 1)		Fitting Y (see figure 1)		Assembly code number 2/
		PIN	Bend angle	PIN	Bend Angle 1/	
1/2 .500 (12.70)	M52471/4-8	M52525/5-8-8	0°	M52525/5-8-8 M52525/5-8-10	0°	A301 A302
				M52525/6-8-8 M52525/6-8-10	45°	A303 A304
				M52525/7-8-8 M52525/7-8-10	90° 3/	A305 A306
				M52525/8-8-8 M52525/8-8-10	90° 4/	A307 A308
		M52525/5-8-10	0°	M52525/5-8-10	0°	A309
				M52525/6-8-8 M52525/6-8-10	45°	A310 A311
				M52525/7-8-8 M52525/7-8-10	90° 3/	A312 A313
				M52525/8-8-8 M52525/8-8-10	90° 4/	A314 A315
		M52525/6-8-8	45°	M52525/6-8-8 M52525/6-8-10	45°	A316 A317
				M52525/7-8-8 M52525/7-8-10	90° 3/	A318 A319
				M52525/8-8-8 M52525/8-8-10	90° 4/	A320 A321
		M52525/6-8-10	45°	M52525/6-8-10	45°	A322
				M52525/7-8-8 M52525/7-8-10	90° 3/	A323 A324
				M52525/8-8-8 M52525/8-8-10	90° 4/	A325 A326
		M52525/7-8-8	90° 3/	M52525/7-8-8 M52525/7-8-10	90° 3/	A327 A328
				M52525/8-8-8 M52525/8-8-10	90° 4/	A329 A330
		M52525/7-8-10	90° 3/	M52525/7-8-10	90° 3/	A331
				M52525/8-8-8 M52525/8-8-10	90° 4/	A332 A333
		M52525/8-8-8	90° 4/	M52525/8-8-8 M52525/8-8-10	90° 4/	A334 A335
		M52525/8-8-10	90° 4/	M52525/8-8-10	90° 4/	A336

1/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
2/ A suffix indicating plating finish shall be specified with the assembly code number, see table VIII.
3/ 90° short drop fitting.
4/ 90° long drop fitting.

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TABLE V. 3/4-inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN	Fitting X (see figure 1)		Fitting Y (see figure 1)		Assembly code number 2/
		PIN	Bend angle 1/	PIN	Bend angle 1/	
3/4 .750 (19.05)	M52471/4-12	M52525/5-12-12	0°	M52525/5-12-12	0°	A501
				M52525/6-12-12	45°	A503
				M52525/7-12-12	90° 3/	A505
				M52525/8-12-12	90° 4/	A507
		M52525/6-12-12	45°	M52525/6-12-12	45°	A516
				M52525/7-12-12	90° 3/	A518
				M52525/8-12-12	90° 4/	A520
				M52525/7-12-12	90° 3/	A527
		M52525/7-12-12	90° 3/	M52525/8-12-12	90° 4/	A529
		M52525/8-12-12	90° 4/	M52525/8-12-12	90° 4/	A534

1/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
2/ A suffix indicating plating finish shall be specified with the assembly code number, see table VIII.
3/ 90° short drop fitting.
4/ 90° long drop fitting.

TABLE VI. 1-inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN	Fitting X (see figure 1)		Fitting Y (see figure 1)		Assembly code number 2/
		PIN	Bend angle 1/	PIN	Bend angle 1/	
1 1.000 (25.40)	M52471/4-16	M52525/5-16-16	0°	M52525/5-16-16	0°	A601
				M52525/6-16-16	45°	A602
				M52525/7-16-16	90° 3/	A603
				M52525/8-16-16	90° 4/	A604
		M52525/6-16-16	45°	M52525/6-16-16	45°	A609
				M52525/7-16-16	90° 3/	A610
				M52525/8-16-16	90° 4/	A611
				M52525/7-16-16	90° 3/	A616
		M52525/7-16-16	90° 3/	M52525/8-16-16	90° 4/	A617
		M52525/8-16-16	90° 4/	M52525/8-16-16	9°0 3/	A622

1/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
2/ A suffix indicating plating finish shall be specified with the assembly code number, see table VIII.
3/ 90° short drop fitting.
4/ 90° long drop fitting.

TABLE VII. 1-1/4 inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN	Fitting X (see figure 1)		Fitting Y (see figure 1)		Assembly code number 2/
		PIN	Bend angle 1/	PIN	Bend angle 1/	
1-1/4 1.125 (28.58)	M52471/4-20	M52525/5-20-20	0°	M52525/5-20-20	0°	A701

1/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
2/ A suffix indicating plating finish shall be specified with the assembly code number , see table VIII.

Fittings materials and finishes shall be in accordance with MIL-DTL-52525, material and plating designators are specified in table VIII.

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TABLE VIII. Material and finish identification codes. 1/ 2/

PIN code material/plating finish	Material	Plating finish
A	Aluminum alloy 6061 or 7075	Anodize in accordance with MIL-A-8625, type II.
AN		Anodize in accordance with MIL-A-8625, type II and NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type II, class 1A. TCP Applied after anodizing.
BN		Bare aluminum with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
C	Steel 2/	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 8/
YC	Chrome-molybdenum steel alloy 4130	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 8/
G	Steel 2/	Zinc plating with colorless passivate in accordance with ASTM B633, type VI, Fe/Zn 8. 3/
YG	Chrome-molybdenum steel alloy 4130	
H	Steel 2/	Zinc phosphate finish in accordance MIL-DTL-16232, type Z, class1.
YH	Chrome-molybdenum steel alloy 4130	
J	Steel 2/	Zinc plating with a colored chromate conversion in accordance with ASTM B633; type II, Fe/Zn 8, or ASTM B695, type II, class 8. 3/ 5/
YJ	Chrome-molybdenum steel alloy 4130	
K	Nickel-copper alloy UNS N04400	No additional finish.
M	High-chromium nickel alloy UNS N06690	No additional finish.
N	Steel 2/	Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 4/
YA	Chrome-molybdenum steel alloy 4130	
YN		Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 4/ 7/.
P	Steel 2/	Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 8. 3/
YP	Chrome-molybdenum steel alloy 4130	
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488, type 2. 6/ 9/
TF		Fluoride phosphate in accordance with SAE-AMS2486. 6/ 9/
U	Steel 2/	Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 4/
V		Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 4/ 7/
W	Steel 2/	Any zinc plating above. 3/ 5/
YW	Chrome-molybdenum steel alloy 4130	
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8. 3/ 5/

1/ Embrittlement test need not be run.
2/ All steels except 4130.
3/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.
4/ Hexavalent chromium free. Finish shall be RoHS compliant.
5/ These finishes have a chromate conversion coating or chromic acid rinse which includes hexavalent chromium and are not recommended.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.
7/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
8/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.
9/ Titanium shall not be used in oxygen or potable water systems.

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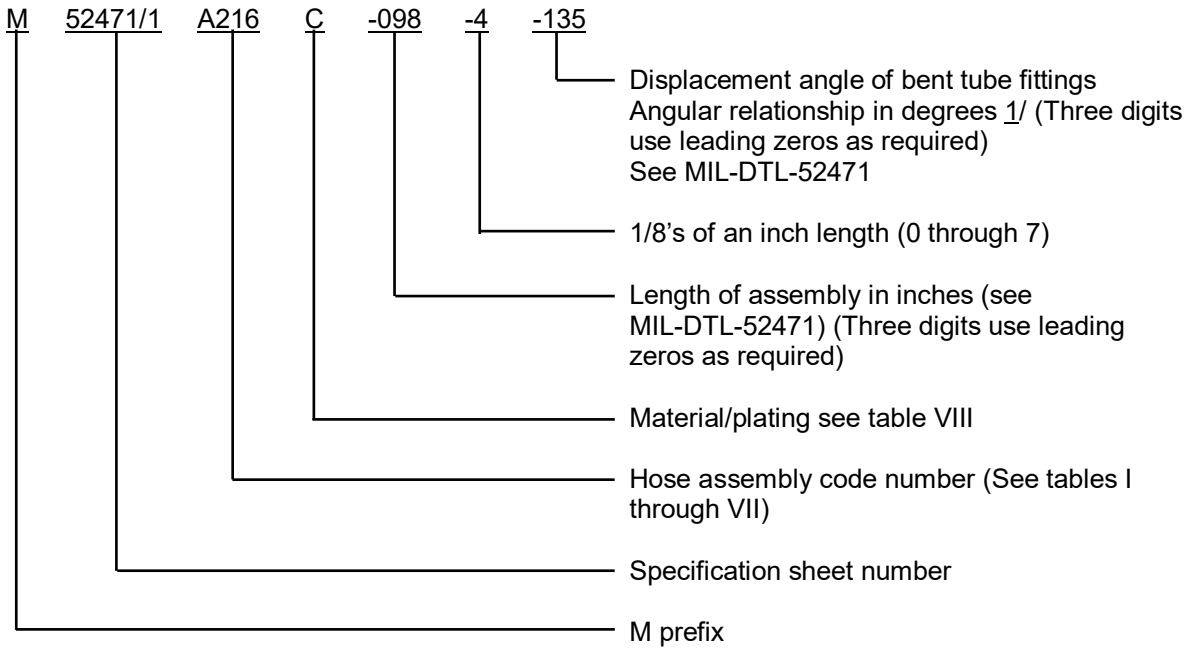
REQUIREMENTS:

The hose assemblies shall be as specified on figure 1 and in tables I through VIII.

Hose shall be in accordance with MIL-DTL-52471/4.

Connector fittings for hose assemblies shall be in accordance with MIL-DTL-52525.

PIN: The PIN consists of the letter “M” the specification sheet number, an assigned code number for the hose assembly, material/plating, hose assembly length, and angular relationship of fittings, when applicable.



1/ Not applicable for assemblies with straight fittings.

PIN example: M52471/1-A216C-098-4-135, describes a hose assembly consisting of M52471/4-6 hose, fitting X M52525/5-6-6, bend angle 0°, fitting Y M52525/6-6-6, bend angle 45°, steel with cadmium plating, 98 ½ inches long, and a displacement angle of 135°.

This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit and function requirements of their application.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

Class I and II ozone depleting substances (ODS) shall not be used in MIL-DTL-52471/1 or any referenced procedures.

Referenced documents shall be of the issue in effect on date of invitations for bid.

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Referenced documents. In addition to MIL-DTL-52471, this document references the following:

MIL-A-8625	SAE-AMS-STD-595/36081	SAE-AMS-STD-595/36270
MIL-DTL-16232	SAE-AMS-STD-595/36099	SAE-AMS-STD-595/36280
MIL-DTL-52471/4	SAE-AMS-STD-595/36118	SAE-AMS-STD-595/36293
MIL-DTL-52525	SAE-AMS-STD-595/36134	SAE-AMS-C-81562
MIL-DTL-81706	SAE-AMS-STD-595/36152	SAE-AMS-QQ-P-416
ASTM B633	SAE-AMS-STD-595/36170	SAE-AMS2417
ASTM B695	SAE-AMS-STD-595/36173	SAE-AMS2488
ASTM F1136/F1136M	SAE-AMS-STD-595/36176	SAE-AMS2486
ASTM MNL37	SAE-AMS-STD-595/36231	SAE-AMS2700
SAE-AMS-STD-595/36076	SAE-AMS-STD-595/36251	T9070-AL-DPC-020/077-2

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2019-030)

Review activities:
Army - AR, AV, GL
Navy - AS, SA

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